A³ WORKSHOP @ MICRO 2026 · CHIA HACKATHON

From Mined Fixes to Gated Repair: Closed-Loop Defect Synthesis for CIRCT

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Code, corpus, every verdict

An agent that reads past compiler fixes wrote inputs that exposed fifteen new CIRCT bugs.

Mutating the same tests found 2, from 150 times as many inputs.

New: reproduced on a second worker, reduced, and matching no known issue. None is filed or maintainer-confirmed yet.

15 new bugs
reading fixes 758 inputs

2 **mutating tests** 115,869 inputs

1 Bugs nobody reports

CIRCT is an open-source compiler for hardware, built on MLIR [1]. It lowers a circuit in FIRRTL or SystemVerilog to Verilog. A crash stops the designer; wrong Verilog can reach a chip.

CHIA [2], the agent framework this hackathon builds on, already repairs reported CIRCT issues. Its bottleneck is supply. Bug-labelled issues:



The open ones have a median age of 911 days.

2 A fix points at its siblings

A past fix corrects one wrong assumption at one site; the same assumption often survives at sibling sites. The agent reads a fix and its test, names the root-cause class, and writes inputs aimed at the siblings. The idea follows AFuzz [4]. New here: tools, not the model, give every verdict; assertions restored; a pre-registered baseline; all as CHIA nodes.

Seed: an August 2026 fix to an out-of-bounds `moore.extract`, the SystemVerilog slice.

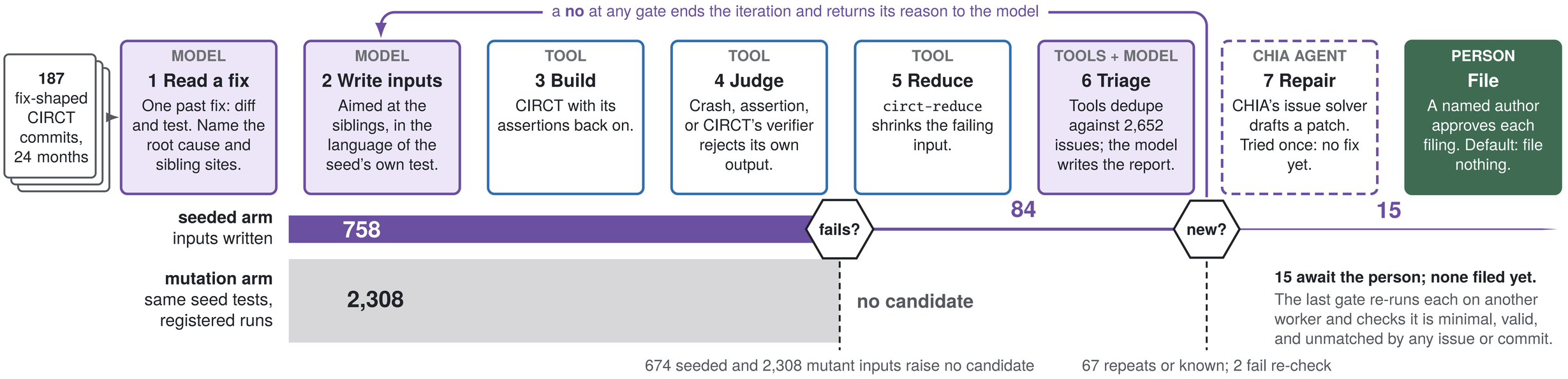
The loop wrote, after reduction (function wrapper omitted):

```
%0 = moore.array_create %arg0, %arg0 : !moore.i32, !moore.i32 -> array<2 x i32>
%1 = moore.extract %0 from -2147483648 : array<2 x i32> -> array<4 x i32>
```

CIRCT (`circt-opt --convert-moore-to-core`) stops:

```
HWOps.cpp:1995: ArrayConcatOp::build: Assertion
`!values.empty() && "Cannot build array of zero elements"' failed.
```

3 The loop: the model writes, tools decide



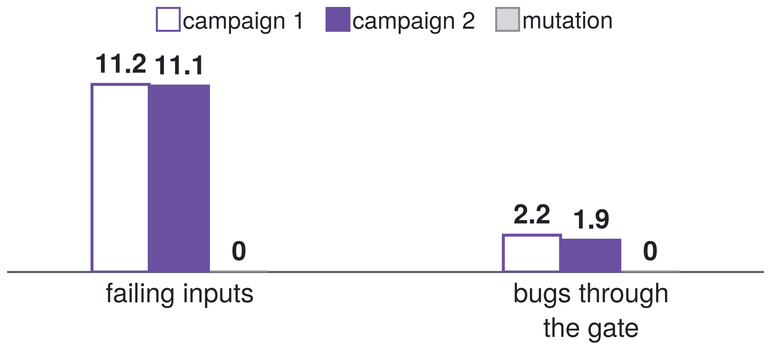
One model (Gemini 3.8 Flash, through CHIA) acts only in MODEL stages; tools answer every gate. Ribbon width is proportional to count, both campaigns summed; the 15 is drawn at a minimum width.

4 A fair test

- Both arms share harness, oracles, reducer and gate; only the generator differs.
- Mutation arm: seed tests rewritten without the fix, after Mut4All [3]; 1,154 mutants from 96 seeds per campaign, no model.
- Registered runs: the mutation arm used up its mutants in minutes. Given the whole window later, it gated 2 bugs (12 vs 2 over ~12 h, different hardware).
- Dedup errs on 1 of 11 distinct and 1 of 11 duplicate labelled pairs.

5 A consistent rate

Per 100 inputs, campaigns about five times apart in size agree. Fifteen bugs show a consistent rate, not an effect size.



Campaign 1: 9 seeds, 134 inputs. Campaign 2: 43 seeds, 624 inputs. USD 14.8 and 10.2 per gated bug at list price.

6 Fifteen new bugs

Three from the seed in box 2; twelve across the compiler, three of them not assertions at all. Examples:

- zero-element array: assertion at `HWOps.cpp:1995`
- `comb.extract`, `llhd.sig.extract` emitted with aggregate types; CIRCT's own verifier rejects them
- segfault in `resolveClassStructBody`, `MooreToCore.cpp`
- abort in `ElaboratedObjectOp::verifySymbolUses`, `OMOps.cpp`

7 What it took

Rebuild the past in minutes: prebuilt LLVM SDKs match the pin of 171 of 187 seeds, so no hours-long LLVM build.

Let the compiler complain: CHIA's image compiles CIRCT's assertions out. `-UNDEBUG` restores them in every object that asserts (536 of 555); the same 1,119 tests pass.

8 Not claimed

- No maintainer has confirmed these bugs; none is filed yet.
- 43 of 187 seeds ran, at 3.6 seeds an hour on Vertex.
- No fix: the repair chain's one patch was a debug print.
- Wrong-output bugs are only reported.

9 Take it

The loop as CHIA nodes and tools, with 736 tests; the corpus; every input, verdict, reduced case and transcript. Prepared for CHIA upstream, none merged yet: the assertions-on image and its build.

`github.com/DevAbhinav-23/CHIA_Hackathon_2k26`

We thank the CHIA developers and the A³ organisers. Claude (Anthropic) helped draft text, code and this poster; the authors answer for every claim.

[1] C. Lattner et al., "MLIR," CGO 2021. [2] A. Cui et al., "CHIA," arXiv:2606.27350, 2026. [3] B. Wang et al., "Mut4All," arXiv:2507.19275, 2025. [4] J. Park, I. Yun, "Agentic fuzzing," arXiv:2605.10074, 2026.